TO 33kV SWITCHBOARD F00-SW-010 PANEL B20
REFER 2000-F00-SLD-EL-10005

F16-NR-011
FLOTATION TRANSFORMER F16-TR-011
NEUTRAL EARTHING CUBICLE

INTERTRIPS TO SIGNALS

F16-TR-011
33 /11 kV
10000 kVA
Dyn11
z=8%
KNAN

254Ω +/-5%
30/0.05A 25A FOR 10s
SENSING
ER-15kV

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

F16-SR-001
11kV FLOTATION SWITCHROOM

F16-SW-011
FLOTATION 11kV SWITCHBOARD

NOTE 4

E-SHUT
F16SR001-LCS
NOTE 5
TO GENERATOR INCOMER

TEST
600/1A 2.5VA Cl 0.2
600/1A 2.5VA 5P20
PQM SEL735
PR SE-330AU
EARTH FAULT INDICATION
TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)
UPSTREAM TRIPS
CLOSE
TRIP
1250A
NOTE 2
PR SEL 751
NOTE 3

TO BUS EARTH SWITCH/VT
11kV 1250A 3-E 25kA FOR 3s
110V AC SUPPLY

A07 A08 A09 2000-F16-SCM-EL-10117 A10 A11 A12 2000-F16-SCM-EL-10123

11kV 1250A 3-E 25kA FOR 3s

100E 160E
50/1A 2.5VA 5P20
50/1A 1.5VA 10P10
VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

F16AG426-VS
SCAVENGER FLOTATION CELL 3 VSD
11/3.3kV 500kW

F16AG427-VS
SCAVENGER FLOTATION CELL 4 VSD
11/3.3kV 500kW

F24PW401-VS
MILL WATER PUMP 1 VSD
11/3.3kV 800kW

F24PW402-VS
MILL WATER PUMP 2 VSD
11/3.3kV 800kW

M 3~ RTD
500 kW
F16-AG-426
SCAVENGER FLOTATION CELL 3 AGITATOR
2000-F16-SCM-EL-10113

M 3~ RTD
500 kW
F16-AG-427
SCAVENGER FLOTATION CELL 4 AGITATOR
2000-F16-SCM-EL-10115

M 3~ RTD
800 kW
F24-PW-401
MILL WATER PUMP 1
2000-F16-SCM-EL-10119

M 3~ RTD
800 kW
F24-PW-402
MILL WATER PUMP 2
2000-F16-SCM-EL-10121

11kV SPARE
FULLY EQUIPPED
-

1CONT'D FROM SHEET 2 DRG 2000-F16-SLD-EL-10001
CONT'D ON SHEET 3 DRG 2000-F16-SLD-EL-10003

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/10/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
26Q	TRANSFORMER PROTECTION DEVICES
27	UNDER VOLTAGE
38	BEARING RTD TEMPERATURE
49W	TRANSFORMER WINDING TEMPERATURE
50	INSTANTANEOUS OVERCURRENT
50SG	INSTANTANEOUS OVERCURRENT SENSITIVE GROUND
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
63	TRANSFORMER PRESSURE / BUCCHOLZ
64	EARTH FAULT
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

DRG No.	REFERENCES
2000-F16-DRG-EL-10001	F16-SR-001 11kV FLOTATION SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F16-SLD-EL-10051	F16-SW-011 11kV FLOTATION SWITCHBOARD KEY INTERLOCKING SINGLE LINE DIAGRAM
2000-F16-SLD-EL-10003	F16-SW-011 11kV FLOTATION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 3 OF 3
2000-F16-SLD-EL-10001	F16-SW-011 11kV FLOTATION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 3
2000-F00-SLD-EL-10005	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 4 OF 5
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2	02OCT25	TRANSFORMER DETAILS UPDATED	[redacted]				
1	29AUG25	FAT REDLINES INCORPORATED		[redacted]			
0	22OCT24	APPROVED FOR CONSTRUCTION			[redacted]	[redacted]	[redacted]

NORTHERN STAR RESOURCES LIMITED

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: ROUGHER & SCAVENGER FLOTATION
F16-SW-011 11kV SWITCHBOARD
SINGLE LINE DIAGRAM SHEET 2

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F16-SLD-EL-10002	2

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